



Reflecting on Attitudes Towards Death Through the use of Immersive Virtual Reality Commercial Video Games

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ABSTRACT

Video games can be an invaluable learning tool beyond pure skill acquisition, such as teaching us how to empathize with others or even self-reflecting on basic existential concerns: isolation, freedom, meaninglessness or death. This is further emphasized with the use of immersive technologies and becomes especially relevant when the experience itself is very difficult to replicate, when not impossible, in the real world. On that regard, this paper analyzes the impact of virtual reality (VR) commercial video games on the existential concern of one's own death. Participants ($N = 30$) played one of three games for 15 minutes and the aftermath was examined using questionnaires and the implicit relational assessment procedure (IRAP). Our results show that there is no difference in the game experience, despite the different gameplay. However, IRAP results seem to indicate that players of the action game have a different attitude towards death.

CCS CONCEPTS

• **Human-centered computing** → Human computer interaction (HCI); Interaction paradigms; Virtual reality; Human computer interaction (HCI); Empirical studies in HCI; Human computer interaction (HCI); HCI design and evaluation methods; Laboratory experiments.

KEYWORDS

Virtual reality, video games, immersive player experiences, existential games, serious games, death-evaluation, death-identity, Implicit Relational Assessment Procedure

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1 INTRODUCTION

Digital games and video games offer people a wide range of means of engagement and experiences, which may differ according to genre, type or mechanics. The evolution in graphic quality and more realistic technologies furthers the impact of such experiences [21]. This is invaluable when games are used as a learning tool that goes beyond entertainment, as proven by the relevance of *serious games* [1]. However, video games can be used to teach us lessons that go beyond pure skill or knowledge acquisition, such as help us empathize with others [3] or even reflect on human existential concerns (*existential games*) [6].

One of such existential concerns that goes along with the human condition is death, the tension between the awareness of its inevitability and the wish to continue to exist [32]. Both fascinating and terrifying, death has a great influence at the personal and societal level in the evolution of our mind, culture, morality, and religion [37]. Therefore, learning how to face our own mortality and coping with this idea is inevitable. However, in contrast with other existential concerns (e.g., isolation, freedom, meaninglessness), one's own death cannot be studied experimentally. Even without taking into account the ethical aspects, anyone who would experience it would obviously no longer be able to benefit from the acquired knowledge or to transmit it to others.

In this interesting scenario, video games may prove again promising. A key element of immersive video games is to be able to feel the emotions resulting from dealing with unusual situations, that the player would probably never be able to experience in the real world. On that regard, it has been shown that some aspects of the dying process can be replicated in VR [18], making it somehow possible to study the lessons acquired from dying, and how they would affect us, experimentally, up to some point. Taking this possibility into consideration, this paper presents a study that investigates the impact of using different types of commercial video games in VR to assess attitudes towards death and its usefulness as coping mechanism. Especially, whether their different approaches to death could lead to different results in anxiety and attitude towards death in real life, letting the player reflect and learn lessons about this basic existential concern. This study also contributes to further the development of more complex immersive existential games about death.

The paper is structured as follows. First of all, Sec. 2, provides a brief overview of the related work on regards to the study of VR as an immersive technology and how the scientific study of death (i.e., *thanatology*) is presented in the literature. Sec. 3 presents the

experimental setup and the chosen methodology, and then the results are shown in Sec. 4. A discussion on such results takes place in Sec. 5. Finally, Sec. 6 concludes this paper and provides some ideas on further work.

2 BACKGROUND AND RELATED RESEARCH

This section provides some background on the main topics required for this study, namely on how VR allows to undergo highly immersive experiences and how to scientifically study attitudes toward death from a psychological perspective.

2.1 The immersive properties of Virtual Reality

Academic researchers have studied VR in itself for decades and its use as a tool in various fields such as the study of human and social behaviors [5, 42]. Initially, studies were more concerned about theoretical aspects, such as "presence" (defined by Witmer as the *"subjective experience of being in an environment, even when one is physically situated in another"* [49]). Later they focused on place illusion and plausibility [40], showing that VR can lead to realistic behavior in immersive virtual environments [39]. The interrelation of presence, engagement, and empathy has also been studied in immersive VR [4].

Even though the use of VR technology in video games also dates several decades, it was necessary to wait until the recent release of affordable Head Mounted Displays (HMD), such as the Oculus Rift in 2016, to actually see VR technologies blooming in the video game industry. Therefore, and despite previous academic research in VR, little scientific studies have been carried out to highlight the differences between this kind of immersive video games and more traditional ones, in particular, regarding the players' experience during VR video games [35]. On that regard, it is important to note that there are two different types of VR: non-immersive and immersive VR [41]. While both deal with 3D environments, the first one refers to 2D visual interfaces such as computer screens or projectors, whereas the second refers to fully immersive interfaces such as HMD [20] or CAVE [7].

Focusing on the latter type, VR enables completely immersive experiences where the virtual world totally replaces the real world, movement and interaction becoming of great importance, even producing body ownership illusions [24]. Compared to non-immersive gaming, using VR enables players to visit real or fictional environments and even experience embodiment in a virtual body, leading to a more intense emotional response and a higher perceived sense of presence [35]. These illusions may even induce a realistic behavior of the people, which are affected as if they had been exposed to the scenarios in reality [9]. For example, it has been shown that embodying a virtual body of a different race [29, 36], can lead to changes in implicit attitudes towards others belonging to that race, or that a virtual body of different appearance can produce changes in behavior [23]. These instances reinforce the idea that immersive experiences can be used as very powerful learning scenarios.

The fact is that playing video games (violent or not) induce a complex combination of feelings, thoughts, and behaviors both positive and negative [15]. Furthermore, playing modern video games produces high arousal emotions which has a central role in

health [34]. Nevertheless, not only the intensity of such emotional experiences is important but also its frequency, since it seems that strong negative emotions (e.g., anger, fear) repeated over time are toxic to health [44]. In addition, VR systems have strong potential to increase the emotional experiences of player.

2.2 Death and fear of death

Death has always exercised fascination to human beings, and thus, significant research has been conducted to study how mortality salience influences the attitudes of people. Thanatology (from Greek *thanatos*, "death") is the scientific study of death and the losses it brings as a result. This discipline, which emerged at the beginning of the 20th century, gained momentum during the second half of that same century following the publication of the landmarks "The Meaning of Death" [8] and "The Psychology of Death" [22].

Terror Management Theory (TMT) is one of the main approaches to try to address how individuals deal with the inevitability of their death and its psychological effects. It postulates that the issue of our mortality is a critical and central point that may (even non-consciously) be responsible for our adherence to fundamental value systems and associated behaviors [11, 38].

TMT posits that people protect themselves from death with a shield of general culture and moral values that transcend the self. As a consequence, when people are facing mortality salience (death thoughts) they are more likely to increase their identification with their social groups (like families, friends, etc.), societal norms or nation, which lead them to negating or rejecting alien individuals or groups. Consequences of mortality salience were compared with a wide range of other types of thought (such as thoughts of pain, failure and social exclusion or television) [26], showing that mortality salience influenced worldview defense and self-esteem promotion, independently of the comparison item.

On regards to the conceptualization of death itself and given the difficulties in analyzing it in terms of direct experiences, the Relational Frame Theory (RFT) [13] tries to consider the derived answers. Indeed, RFT has demonstrated much utility in conceptualizing and modeling complex verbal behaviors, especially regarding death and suicide [14].

The implicit association test [12] is generally used to assess implicit attitude, although more recently the Implicit Relational Assessment Procedure (IRAP) [2] proved to be an interesting alternative [47]. Indeed, it has been used to explore implicit attitudes to life and death with two IRAPs [17], one that assessed relational responses specific to death and life with respect to self (i.e., self, others, life, and death), and a second that assessed relational responses specific to evaluations of death and life (i.e., life, death, positive, and negative). As expected, both indicated pro-life biases, but they failed to demonstrate the presence of anti-death biases, and even found in one case a specific death-positive bias. However, a recent study also using two death related IRAPs has shown that suicidal ideation is associated with an implicit "fearlessness of death" [16].

When it is time to link death and play, both are omnipresent, appearing in plenty of events, occasions, and places. According to McAllister and Ethan [31], computer games cultivate a perception that "real" death is predictable, controllable, reasonable, and ultimately benign. It enables computer games to provide mundane,

remarkable, humbling and empowering experiences. Thus, video games have a potential become an ideal method to teach or let the players reflect about our own mortality, teaching important life lessons.

3 METHODOLOGY

This section presents the experimental setup and methodology, the selection process for commercial video games and procedure for the experiment sessions.

3.1 Materials

To run the video games, we used a Windows 10 computer powered by an Intel i7 processor with 16 MB of RAM with one Nvidia GeForce GTX 1070 graphics card connected to an HTC Vive HMD (https://en.wikipedia.org/wiki/HTC_Vive). The video games were run through the Steam platform ([https://en.wikipedia.org/wiki/Steam_\(software\)](https://en.wikipedia.org/wiki/Steam_(software))) and SteamVR, its VR plugin. The HMD and its two controllers were tracked with 6 degrees of freedom in position and rotation using the Lighthouse tracking system supplied with the Vive, in a physical area limited to 3x3m. The headset covers a nominal field of view of about 110° (approximately 90° per eye) through two 1080x1200 pixel displays that are updated at 90Hz. This setup allowed a fully immersive gaming experience.

3.2 Game selection process

In order to choose the appropriate games, three overall core game experiences (categories a, b and c) were defined for the experiment, each one addressing the topic of death from a different perspective.

- **Spooky Game.** An unsettling experience where the player is constantly afraid, looking like something scary is about to happen at every corner, maybe even dying. However, in this experience, it is deemed important that death, should it occur, is not cheapened by becoming a common occurrence and, ultimately, a nuisance to the play experience.
- **Action Game.** A high action experience where players and opponents alike are constantly killing each other. Death is not just a possibility, but an integral part of gameplay, of minor consequence overall. Death should come preferably with no suffering in the process, even looking cartoonish.
- **Surreal Game.** A surreal, trippy, maybe even out-of-body experience, where the life of the player is never at stake. The game concept itself should make it very clear to the player that death is not part of the gameplay at all, focusing mostly on exploring or interacting in a funny game environment disconnected from reality.

To choose a game for each category, some common desirable requirements were set. The goal was to make the base interaction as similar as possible between the games, despite the difference in themes, as well as making it easier for subjects to pick up and understand the gameplay during the experiment session, so they could focus on the experience itself.

- First of all, obviously, the games had to be fully compatible with the chosen hardware requirements.
- Native VR support. The player had to be immersed in a real 3D surround experience, and not through a 3D screen-like

one, such as the method provided by some drivers or mods that adapt non-VR games to VR experiences.

- Play area and positioning requirements had to be the same for all games. The chosen method was standing still (no need to walk within the physical world play area) to minimize the possibility of motion sickness.
- The same input device would be used for all games. The chosen input method was through motion tracker controllers (i.e., no keyboard+mouse or gamepad), in order to provide a more immersive experience.
- Easy and intuitive controls that could be learned on the go, with minimum previous explanations or intervention during the session.
- The possibility to have a meaningful game experience in no more than 15-20 minutes. For instance, being able to complete a full stage, or playing up to some moment when stopping the session would not completely disrupt the game flow (i.e., some kind of obvious breakpoint in gameplay).
- No need to follow a narrative to understand the game or what the player must do at any moment. It should not be important to carefully listen to voice-overs or read subtitles in order to follow the game. Especially, in a language not native to the player.

The selection was conducted through the VR section of Steam digital store [48]. At the moment of this selection process, the store boasted 4030 games with native VR support, among which 3918 supported the hardware of choice (HTC Vive). Once the positioning and input requirements were taken into account, the number was reduced to 2804. In order to further prune the list and search for suitable candidates, games were filtered using different tags such as, for instance, "spooky", "horror", "terror" (category a); "action", "shooter", "FPS" (category b); "surreal", "colorful", "casual" (category c). Extra care was taken during this step, since tags are selected by Steam users themselves, and it has been detected that, in some cases, they are made up for fun or jokes (e.g., "The Lego Movie 2" videogame was tagged at that moment as "Psychological Horror" and "Sexual Content" (<https://bit.ly/3ogP5RU>)).

Using the tag system reduced the list to 202 (category a), 355 (category b) and 261 (category c) games. These remaining games were further studied looking at their descriptions and reviews in the Steam store section (general score and written comments), as well as on external review pages, selecting those which could fulfill the rest of stated requirements and were considered most immersive. This resulted in a total of 19 final candidates among all categories. The final step was to install, test and evaluate each of these videogames individually, through direct gameplay. The main reasons for discarding most candidates during this final stage were VR motion sickness concerns or the inability to provide a meaningful game experience unless they were played for a longer time.

The three chosen games, one for each respective category, were a) "AFFECTED: The Manor", b) "SUPERHOT VR", and c) "GNOG". Following, a brief description copied from the developer description or some user reviews is included, as well as a sample screenshot (see Figure 1).

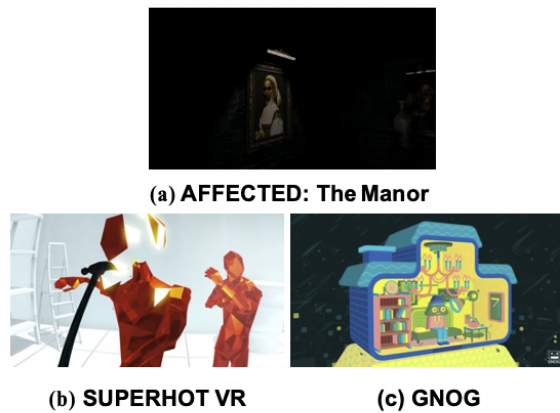


Figure 1: Game screenshots by chosen category

- AFFECTED: The Manor [43] - (Spooky game) - "A well done haunted house experience full of jump scares. Very creepy atmosphere and sounds. If you are looking for a scare or looking to scare someone else, look no further."
- SUPERHOT VR [45] - (Action game) - "The definitive VR FPS in which time moves only when you move. No regenerating health bars. No conveniently placed ammo drops. It's just you, outnumbered and outgunned, grabbing weapons off fallen enemies to shoot, slice, and maneuver through a hurricane of slow-motion bullets".
- GNOG [25] - (Surreal game) - "A 3D puzzle game set in a tactile world of toys and secrets. Point, click, grab, poke, spin, pull, and play with uniquely charming monster heads as you explore the hidden worlds inside them. Filled with eye-catching designs, playful interactions, and a rich, reactive soundtrack, the hand-crafted heads come to life in VR".

3.3 Recruiting

In order to recruit participants, researchers shared a form for registration through the official Twitter account of our institution and its web page. The form included a self-registration system with a pool of dates and times to schedule its participation. As the experiments were performed in pipeline, the reservations took place every 30 minutes, even though the whole process lasted for up to one hour.

3.4 Experiment setup

The participant was invited to come at the institution's premises at the established hour. Researchers took him to the VR room where an informed consent form had to be read and signed before playing. The type of game was assigned to participants randomly through a process respecting the balancing order between participants gender and condition.

The VR part of the experiment lasted approximately 30 minutes including the setup and the VR gaming experience and then another 30 minutes was dedicated to the tests. The time spent playing was actually of exactly 15 minutes, but the player was not informed about it. Researchers offered indications about the game controls

beforehand, depending on the game. Once the game experience was complete, researchers asked participants about their time perception and ask them how long they had been playing. In order to reduce waiting time, once a VR session was complete, the participant was taken to another room for the tests, another VR session could be started with minimum waiting time.

3.5 Participants

Thirty participants (12F, 18M), aged 18 or over, volunteered for the study. The mean age and the standard deviation for females were 34.3 and 10.8, while for males were 36.1 and 8.4. No participant left due to simulator sickness, fear or any other reason, although all participants were informed that they could leave at any time without giving any reason.

Regarding profiles, 10% never (or almost never) play games, 50% play sporadically (they prefer fun or relaxing games rather than games that require a great learning curve or high levels of commitment or participation), 30% play often (they prefer games with a large learning curve, that require a higher level of participation or represent complex tactical challenges), and 10% describe themselves as unconditional (they enjoy high-action, extremely competitive games that require a high level of participation, commitment and skill to progress).

3.6 Measures

Aiming to get different perspectives on the participants experiences, we employed self-reported questionnaires. Measures obtained refer to level of anxiety before/after playing, perception of the time spent playing, game experience, fear of death, and implicit relation between death/life and positive/negative feelings.

Upon arrival, participants completed the 6-item short-form of the state scale of the Spielberger State-Trait Anxiety Inventory (STAI) [30] to estimate the level of anxiety. It consists of 6 questions participants have to rate in a Likert scale from 1 to 4, where 1 is not at all, 2 somewhat, 3 moderately, and 4 very much. This questionnaire was translated into Spanish. The responses are used to compute the total STAI score, which ranges between 20 and 80, being 34-36 a "normal" score.

Just after playing one of the three games for 15 minutes and before removing the HMD, participants were asked to report orally 'what time did you spent playing?' in order to have an estimate of the time they thought they spent playing. Afterwards, participants answered the short form of STAI again, which enables the comparison of the anxiety level before and after playing.

Then, participants answered both the 'in-game' and 'post-game' experience sub-scales from the game experience questionnaire (GEQ) [19]. The in-game version of the GEQ (which is a concise version of the core module) assesses game experience as scores on seven components: immersion, flow, competence, positive and negative affect, tension, and challenge. Similarly, the post-game module assesses how players felt after they stopped playing using four components: positive experience, negative experience, tiredness, and returning to reality. The items and answers were also translated into Spanish. The score for each component is the mean

of the answers to the related questions. For each question, participants have to rate in a Likert scale from 1 to 5, where 1 is not at all, 2 slightly, 3 moderately, 4 fairly, and 5 extremely.

In order to study the fear of death, we used both the 'self' and 'other' sub-scales of the Spanish adaptation of the Collett-Lester Fear of Death Scale [28, 46]. Each sub-scale is composed of 7 items. For each question, participants have to rate in a Likert scale from 1 to 5, where 1 is not, 3 somewhat, and 5 very. Thus, the minimum possible score is 7 and the maximum possible is 35.

Finally, we used an objective measurement of participants attitude toward death based on the Implicit Relational Assessment Procedure (IRAP) [2] relying on the GO-IRAP software [33]. In this procedure, participants must accurately and quickly respond to the relation between two stimuli presented on screen using one of two response options. In our experiment, all the stimuli used in [17] were translated to the subject's native language.

Across pairs of blocks, participants must respond using two contrasting response patterns, e.g., "Life-Positive option" vs "Life-Negative option". Afterwards, the reaction times are compared between these blocks. The basic hypothesis of the procedure is that average response latencies should be shorter across blocks of consistent relative to inconsistent trials. Thus, it is expected that participants respond more rapidly to relational tasks that reflect their current beliefs than to tasks that do not.

Following the instructions detailed in [27] the response latencies from each participant were converted into D-IRAP Score. These scores have a theoretical range from -2 to +2. Positive score indicates a bias for consistent block types, while negative scores indicates bias for inconsistent block types.

4 RESULTS

This section summarizes the results obtained for each measure using graphics and statistical tests. Normality was assessed through the Shapiro test.

Anxiety - Figure 2 shows boxplots of the STAI scores before and after playing each videogame.

The median of the scores slightly increased while playing for participants assigned to the spooky game (from 30 to 35). However, there was one of these participants who started with a relatively high level of anxiety. The dependent 2-group Wilcoxon Signed Rank test was applied to compare the STAI scores before and after playing and assess whether their population mean ranks differ. $V = 11.5$, $p\text{-value} = 0.400$. The median of the scores increased more for participants assigned to the action game (from 38.5 to 45). $V = 9.5$, $p\text{-value} = 0.261$. On the contrary, the median of the scores for participants playing the surreal game decreased a bit (from 28.5 to 28). $V = 26$, $p\text{-value} = 0.720$.

Time perception - Boxplots in Figure 3 describe the time evaluation of participants according to each video game. The range is 10 to 30. The median matches the true value (i.e., 15 minutes) for the groups assigned to the spooky and the surreal games. Participants playing the action game overestimated the time they played. The Kruskal-Wallis one-way test was applied to test whether samples originate from the same distribution. $H = 4.398$, $p\text{-value} = 0.111$.

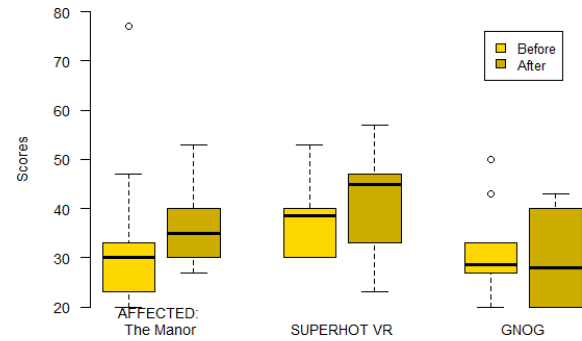


Figure 2: STAI scores before and after playing

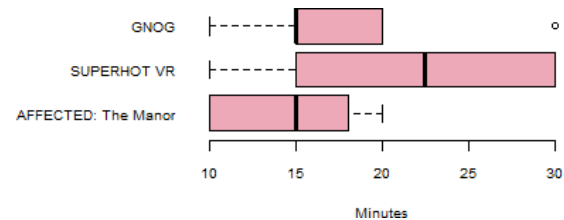


Figure 3: Time perception

In-game experience - The distributions of the scores for the 7 components and the 3 games are shown in Figure 4. In addition, the results of Kruskal-Wallis one-way tests are detailed.

- **Competence:** The highest median is related to the spooky game. All games have a similar range. $H = 0.839$, $p\text{-value} = 0.658$.
- **Sensory and imaginative immersion:** The highest quartiles are those of the spooky game. The action game has a large range. $H = 4.379$, $p\text{-value} = 0.112$.
- **Flow:** All games get a relatively high score, but the highest values belong to the surreal game and the lowest to the action one. $H = 0.828$, $p\text{-value} = 0.661$.
- **Tension:** In general, there are low scores for this component. The action game has the highest 3rd quartile. $H = 0.468$, $p\text{-value} = 0.791$.
- **Challenge:** All groups present a high median. However, the action game tends to have higher values while the scores of the spooky game are highly variable. $H = 4.692$, $p\text{-value} = 0.096$.
- **Negative affect:** The scores are low, in particular for the action and surreal games. $H = 0.628$, $p\text{-value} = 0.731$.
- **Positive affect:** The scores are in general high. The distributions for the action and surreal games are similar. $H = 6.901$, $p\text{-value} = 0.032$.

Post-game experience - The distributions of the scores for the 4 components and the 3 games are shown in Figure 5.

- **Positive experience:** The three boxplots show a relatively high variability. Many of the highest scores are those of the action game. $H = 2.230$, $p\text{-value} = 0.328$.

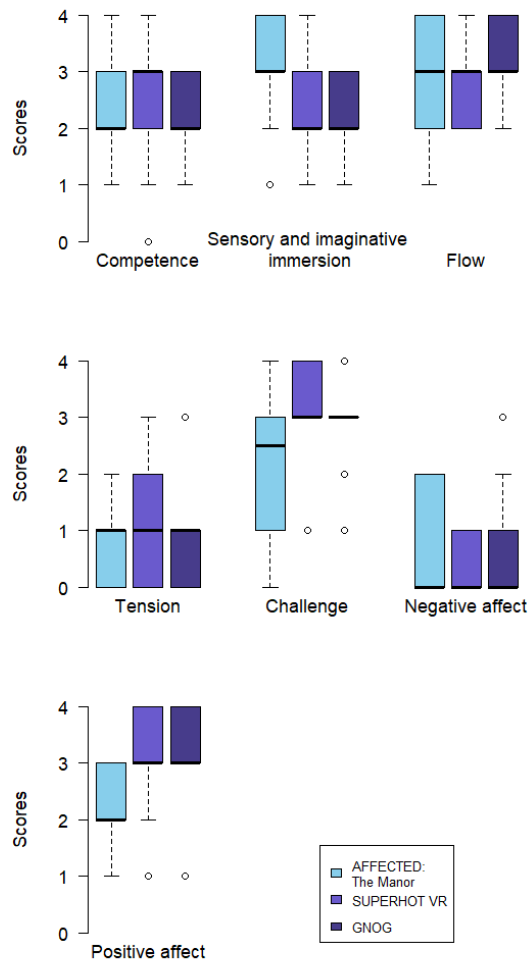


Figure 4: In-game experience scores

- *Negative experience*: The scores show low variability, and all the quartiles are low. All games present relatively similar distributions. $H = 2.207$, $p\text{-value} = 0.332$.
- *Tiredness*: Most scores are between 0 and 2. The lowest score are those of the surreal game. $H = 2.113$, $p\text{-value} = 0.348$.
- *Returning to reality*: Most scores are lower than 1. However, the highest scores belong to the spooky game. $H = 2.611$, $p\text{-value} = 0.271$.

Fear of death - The distributions of the scores for the 2 components and the 3 games are shown in Figure 6.

- *Self*: The highest median belongs to the action game, but the differences between medians are relatively small. There is a relatively high variability in all the distributions. $H = 1.093$, $p\text{-value} = 0.579$.
- *Other*: The medians are very similar, but the third quartile of the spooky and action games are higher than that of the surreal game. $H = 0.683$, $p\text{-value} = 0.711$.

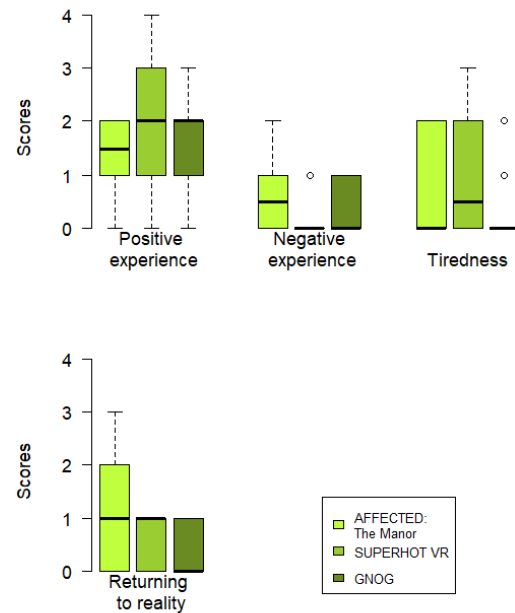


Figure 5: Post-game experience scores

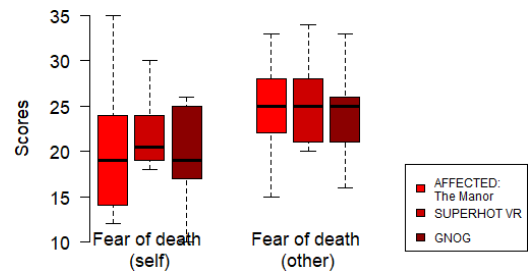


Figure 6: Fear of death scores

IRAP - Figure 7 shows the D-IRAP scores (mean and standard error of the mean) for each pattern and game. The one-way analysis of variance is applied for each pattern to compare population means among games.

- *Life positive*: The mean scores are positive and relatively different, while the standard errors are small. Thus, participants have a positive bias for that pattern. $F = 4.541$, $p\text{-value} = 0.022$.
- *Life negative*: The mean scores are also positive but smaller than those of the previous pattern. The differences among means decrease. $F = 1.062$, $p\text{-value} = 0.362$.
- *Death positive*: The mean scores are negative but small in absolute values. The smallest mean score is that of the spooky game. Thus, participants have a relatively small bias for that pattern. $F = 0.545$, $p\text{-value} = 0.587$.
- *Death negative*: The mean scores for the spooky and the surreal games are positive, but the second is very small. In

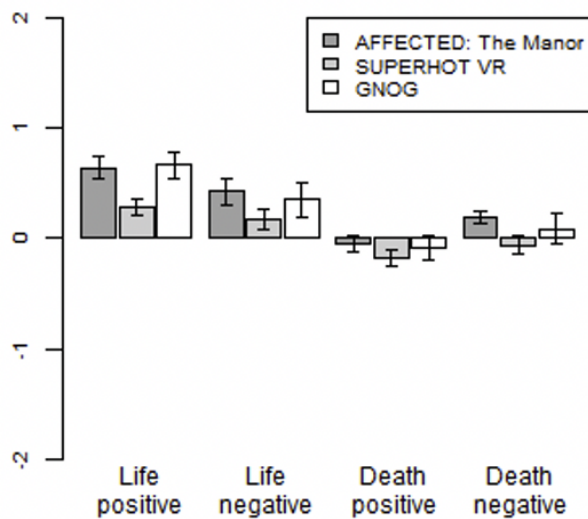


Figure 7: D-IRAP scores

contrast, the mean score is small and negative for the action game. $F = 1.494$, $p\text{-value} = 0.246$.

5 DISCUSSION

This paper explores the use of immersive VR video games as a method to reflect on the existential fear of death and learn about how the experience affects the player. For that purpose, we selected video games with comparable VR modalities, but very different gameplays and evocation of death. We let the participants play one of the three games in VR for 15 minutes, and then we examined the aftermaths using both questionnaires and the implicit relational assessment procedure.

Although the chosen games and their gameplay were very different, our results showed a similar gaming experience for the three games. We observed important variability, but there was no game showing a result significantly different from the other. Especially we can remark that the negative affect and negative experience were very low, whereas the positive affect was relatively high.

The STAI questionnaire responses showed that there was no significant difference regarding the anxiety level of participants before and after playing. This is important because this shows that despite the games were different the gaming experience was similar. However, the time perception seemed to be different and players of the action game have shown higher estimation of their played time, even if there was no significant difference. Concerning the Fear of death, action game players presented the highest median for the Self sub-scale whereas that players of the spooky and action games presented the highest scores in the Other sub-scale, even if there was no significant difference between the three games.

However, the IRAP D-Scores results showed a significant difference regarding the association of Life-Positive, where the action game's players presented the lowest D-score. This is interesting because it seems to indicate that players of the action game, that

were often confronted to death, have a different attitude toward life, since they seemed to agree less strongly than the other players about the positiveness of life.

The D-Scores for Life-Negative and Death-Positive, both are relatively small and similar for the three games. Player seemed not to be in contradiction with the idea that life is negative, whereas they seemed to disagree with the idea that Death is Positive. This is not so surprising, since life not only has positive aspect, but also negative ones, and death is usually not considered positive.

Remarkably, although the last block, Death-Negative did not show a significant difference between the three games, we can observe that both the spooky and surreal games present positive values, indicating that participant agree with the proposed assertion, however it is negative in the action game showing that the players seemed to disagree with the assertion. This is also interesting as players of the action game (confronted to death), seem to have a different attitude toward death. Indeed, they seemed in contradiction about the negativeness of death.

The main limitation of our experiment concerns the fact that participants played one game only one time for 15 min. Nevertheless, most video games are played repeatedly. It would be therefore interesting to have repeated game sessions, to be closer to a usual gamer behavior.

6 CONCLUSIONS

This paper presented an experiment that assessed how playing different commercial video games in virtual reality (VR) during a short time could be used to reflect and learn about one of the basic human existential concerns, one's own death. On that regard, many studies involving VR games use games specially developed for experimental purpose, such as [10] and are not always implemented and tested as a full-scale game, reaching mainstream players.

Measurements were based on the effect of time perception, anxiety and attitude toward death and life. In particular, three games were chosen, which address the topic of death from a different perspective: a) an unsettling experience where something scary (or dying) is about to happen; b) high action were player and opponent are constantly killing each other; and c) a surreal game, where the life of the player is never at stake, but akin to an out-of-body experience. Our results show that playing any of the games did not have a statistically significant effect on the time perception, anxiety (measured by STAI scores), in-game and out-game experience scores, or fear of death scores. However, the descriptive analysis of the samples suggests that participants playing the action game are more inclined to lose track of time. In addition, IRAP results seem to indicate that players of the action game, which were often confronted to death, show a different attitude toward life and death, providing directions on how to design an existential game based on these assumptions.

It still seems necessary to study commercial VR video games and their effects, on players' feelings, thoughts, and behaviors, in the short and longer term and several lines of research stem from this work. In this work we have focused on three very different perspectives for addressing the topic of death, but other perspectives could be explored (e.g., games where death happens, and it has a negative consequence for the player such as losing points). Furthermore, it

would be interesting to analyze the longitudinal effect of playing death-related games periodically, which could be assessed by programming more sessions with a longer duration. Another research line could be to analyze whether there are significant effects for specific segments of the population (i.e., study in detail factors such as gender, age, experience with immersive technologies, etc.).

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